



Interface Design

Ing. Lisa-Maria Freller, MA

Responsive Design

Today

- 📌 Responsive Design
- 📌 Mobile Design Patterns
- 📌 UX management



🎯 **Goal:** Understand responsive Design, use mobile Design Patterns

Challenge #5

Responsive Redesign
2 people per team



Use **Figma Sites** or **Figma** to design a mobile responsive Design for your desired topic

Create layouts for different devices and submit Figma file including:



- Mobile (360 px) and Mobile Large (430 px)
- Tablet (768 px)
- Desktop (1440 px)



Consider following:

- Design Patterns may differ (eg. Breadcrumbs)
- Beware touch interaction states and minimum touch target sizes
- The components & design patterns differ in mobile context
- Include layout changes: Single col on mobile, Two columns on tablet, Multi-column on desktop - document your grid

Responsive Design

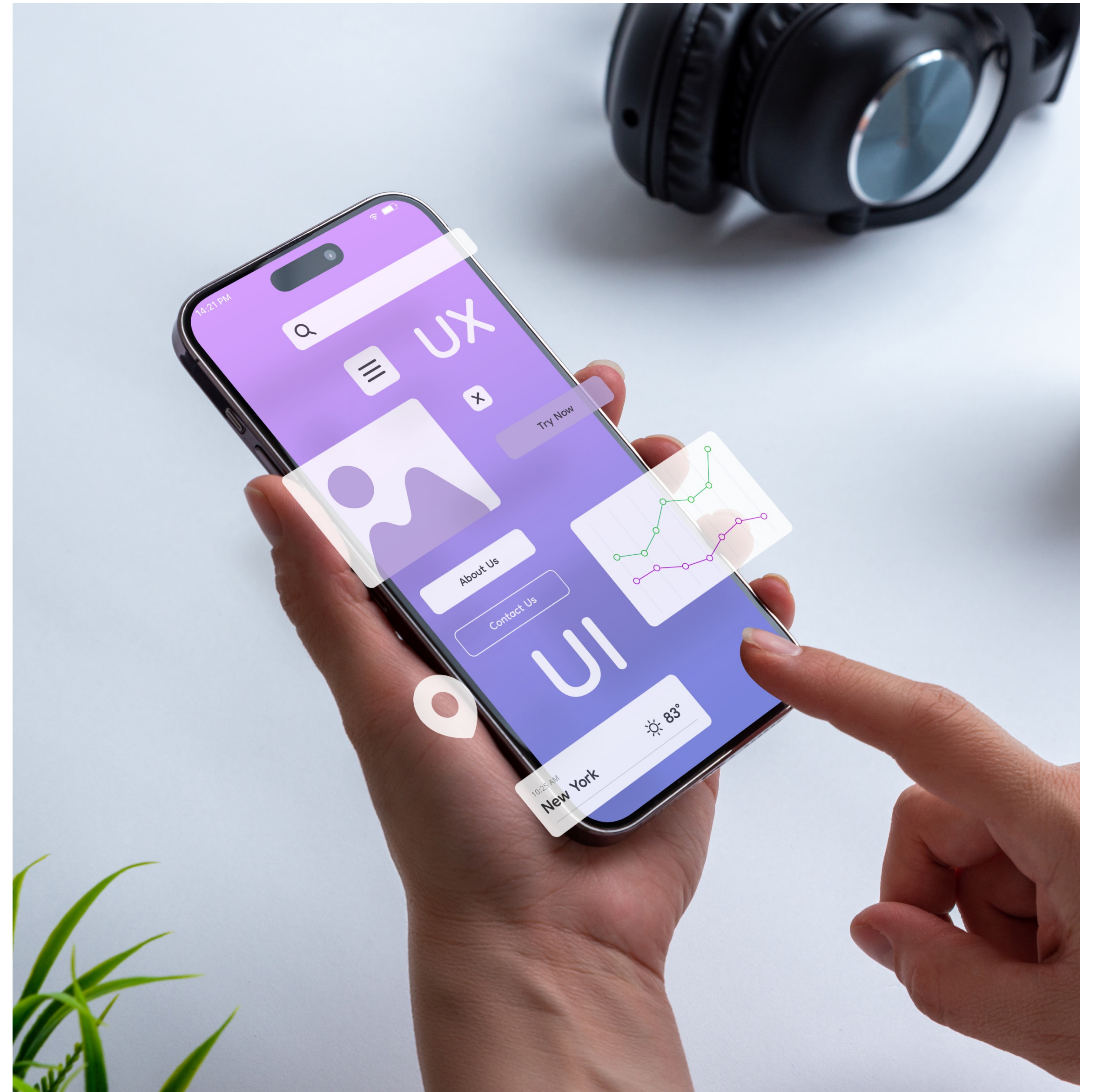
Responsive Design means that designs adjust on tablets, phones, and desktops of any resolution



Responsive Design

Layout, images, and content automatically adapts to different screen sizes and devices

- ✓ **Mobile-first design** means beginning with the mobile experience and progressively enhancing for larger screens. The constraint of a small screen forces ruthless **prioritisation**. Less space means more clarity needed and the importance of elements has to be obvious.
- ✓ **Touch vs. Mouse interaction**: Touch needs bigger click targets, enough padding, easy navigation



Mobile <> Desktop

Layout, Grids, Breakpoints

States

Touch-Friendly Design

Navigation & Design Patterns

Security



Last login
Feb 16, 2026 at 09:15 AM



Password changed
Jan 15, 2026

Change password ▾

New password

.....

New password

.....

Forgot password

Confirm password change

Security



Last login
Feb 16, 2026 at 09:15 AM



Password changed
Jan 15, 2026

Change password ▾

Password

.....

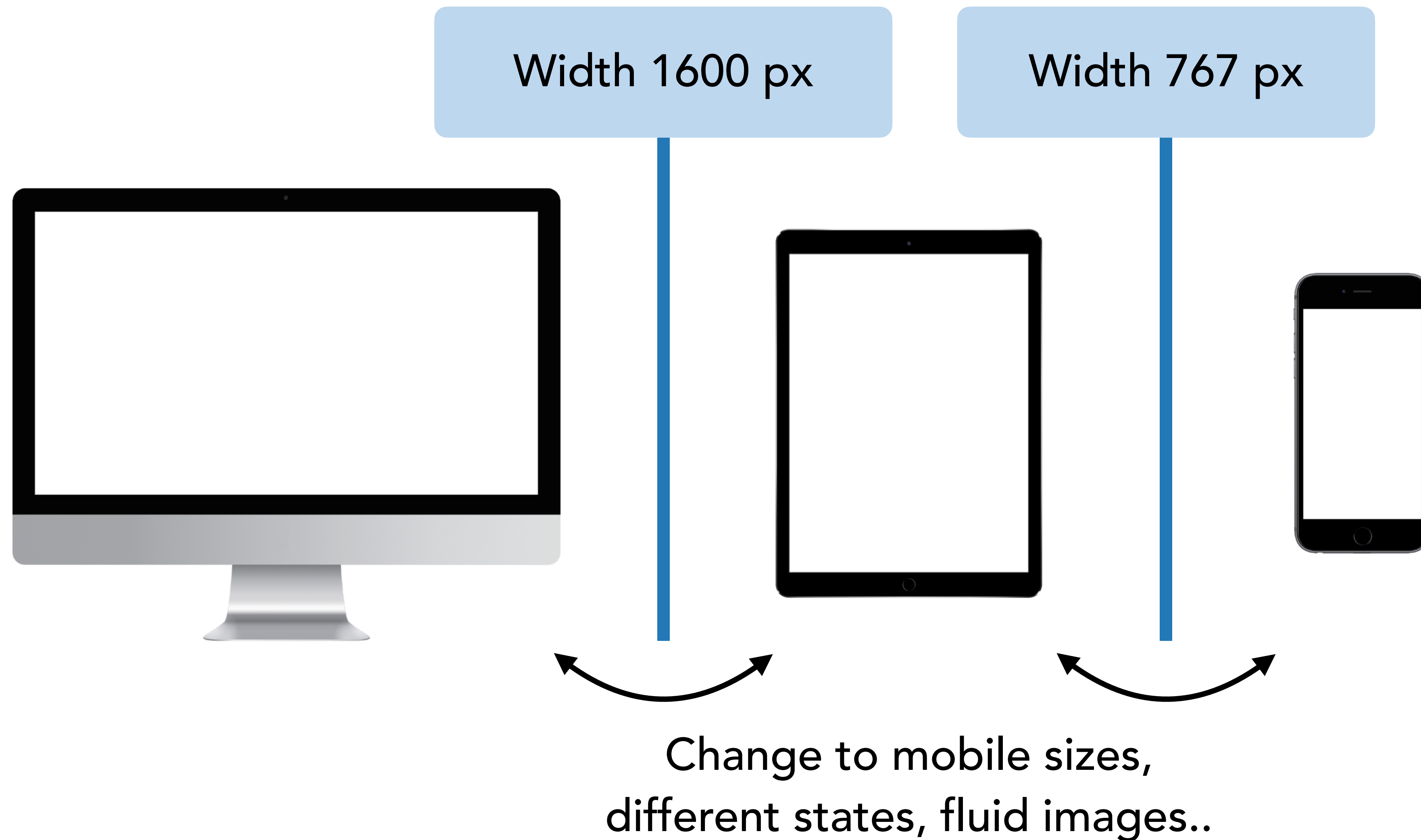
New password

.....

Forgot password

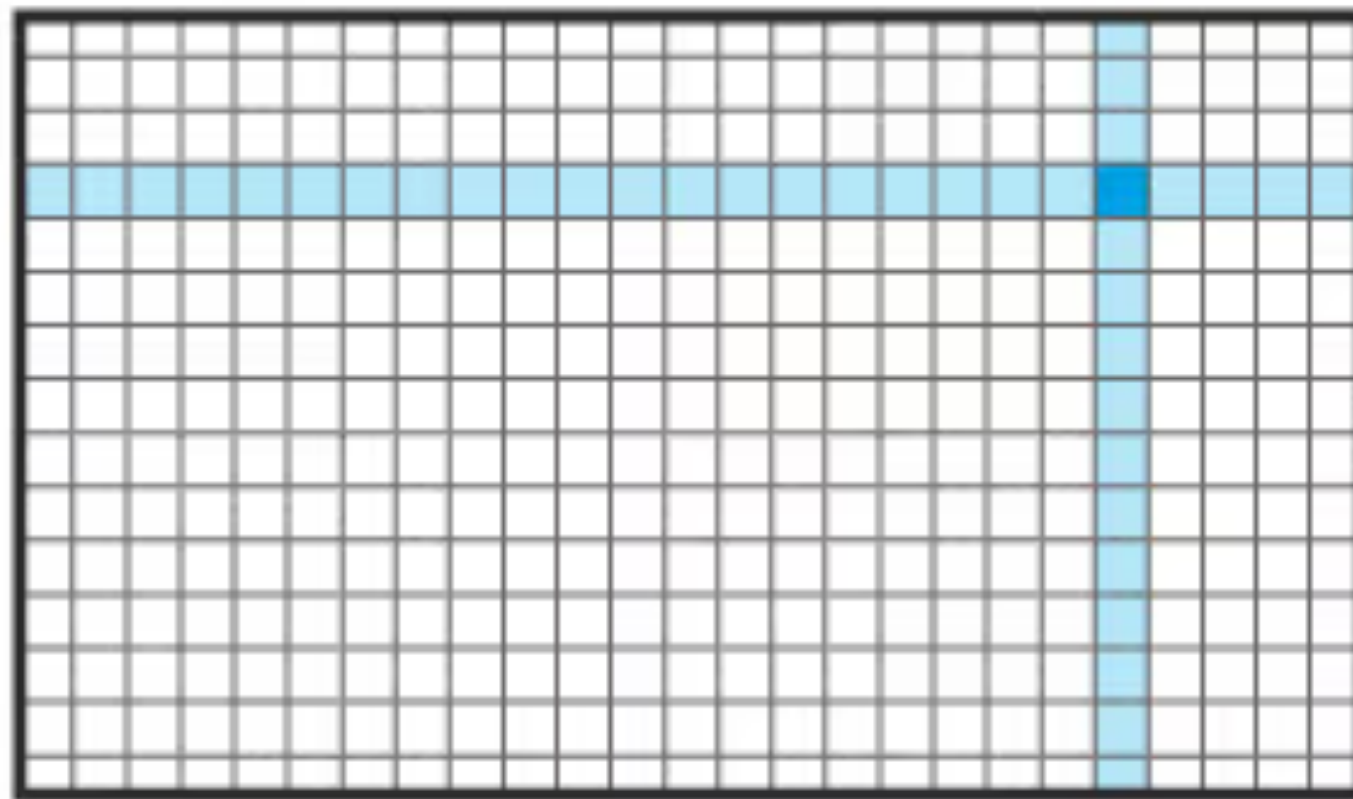
Confirm password change

Breakpoints

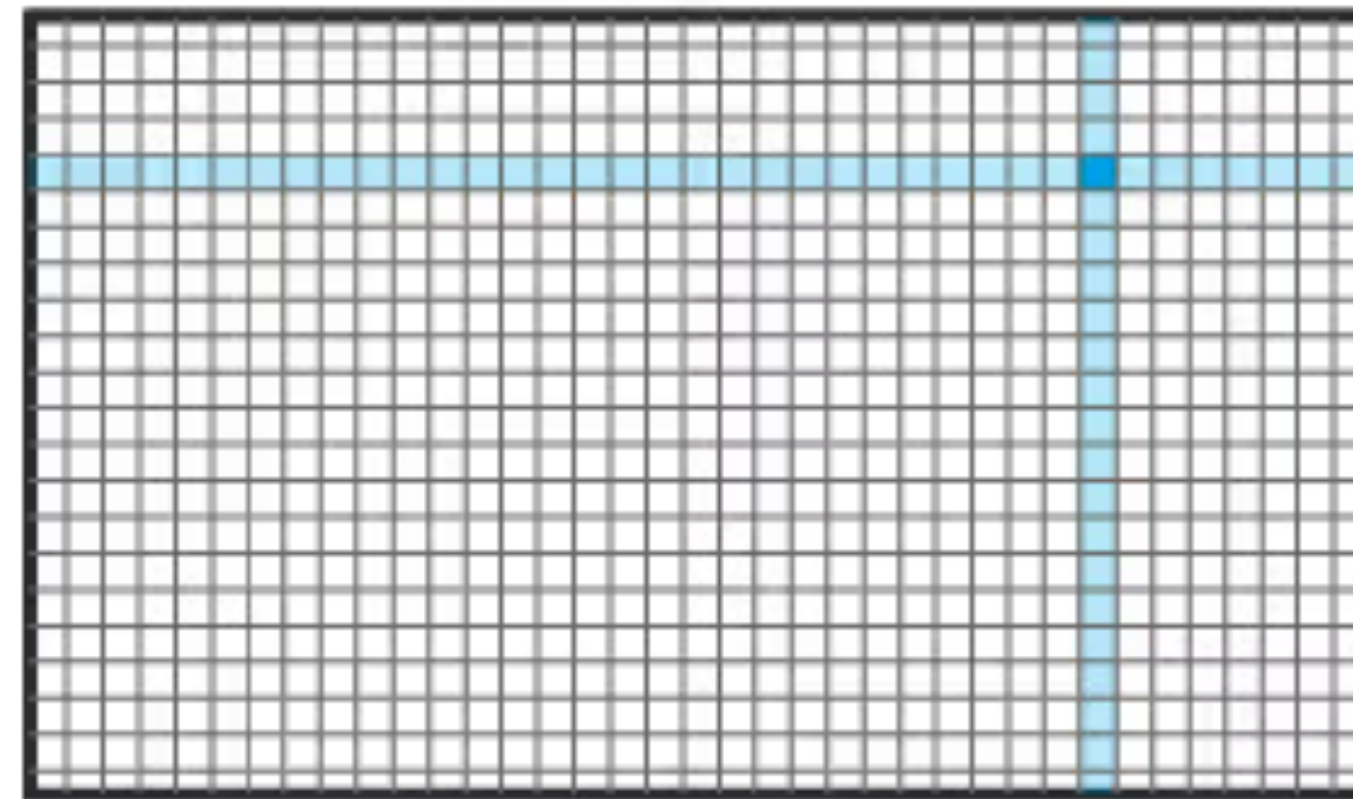


Grid

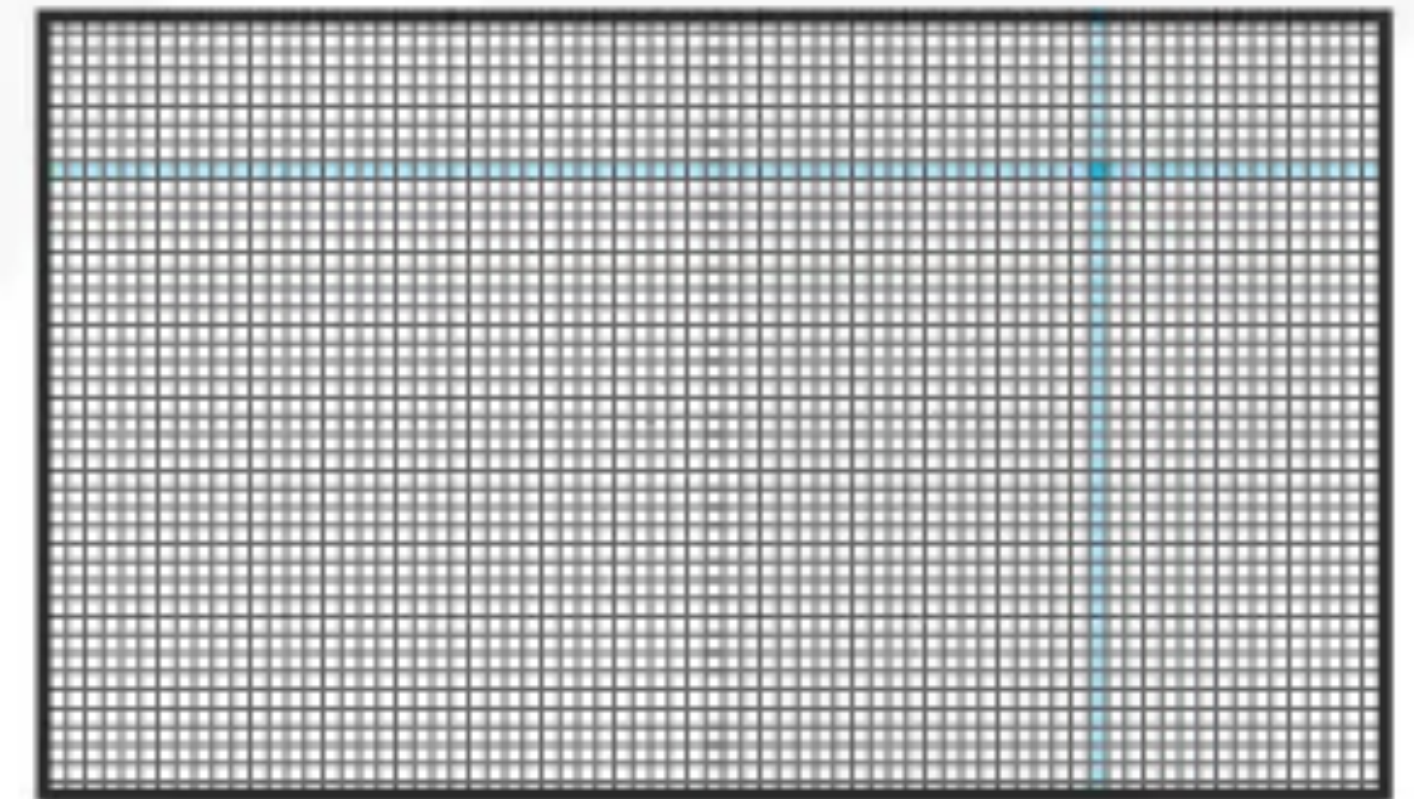
720p (HD)
1280 x 720 pixels



1080p (Full HD)
1920 x 1080 pixels

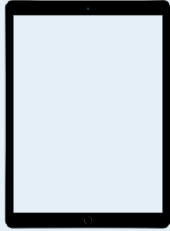
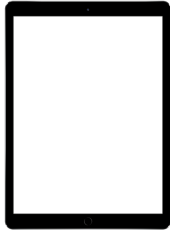


4K (Ultra HD)
3840 x 2160 pixels

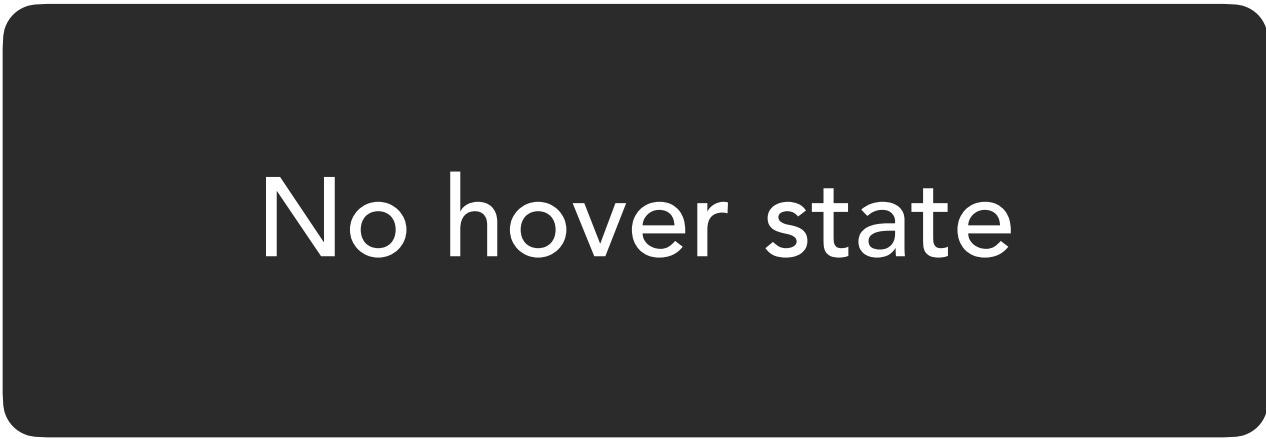


The absolute size method doesn't work because devices vary in size.
Therefore, responsive design uses **relative sizes** for optimal **space usage**.

Grid

	BREAKPOINT	COLUMNS	MARGIN	GUTTER
	> 1600 px	16	20	24
	1599-1200px	12	20	24
	1199-769px	8	24	24
	<767px	4	24	24

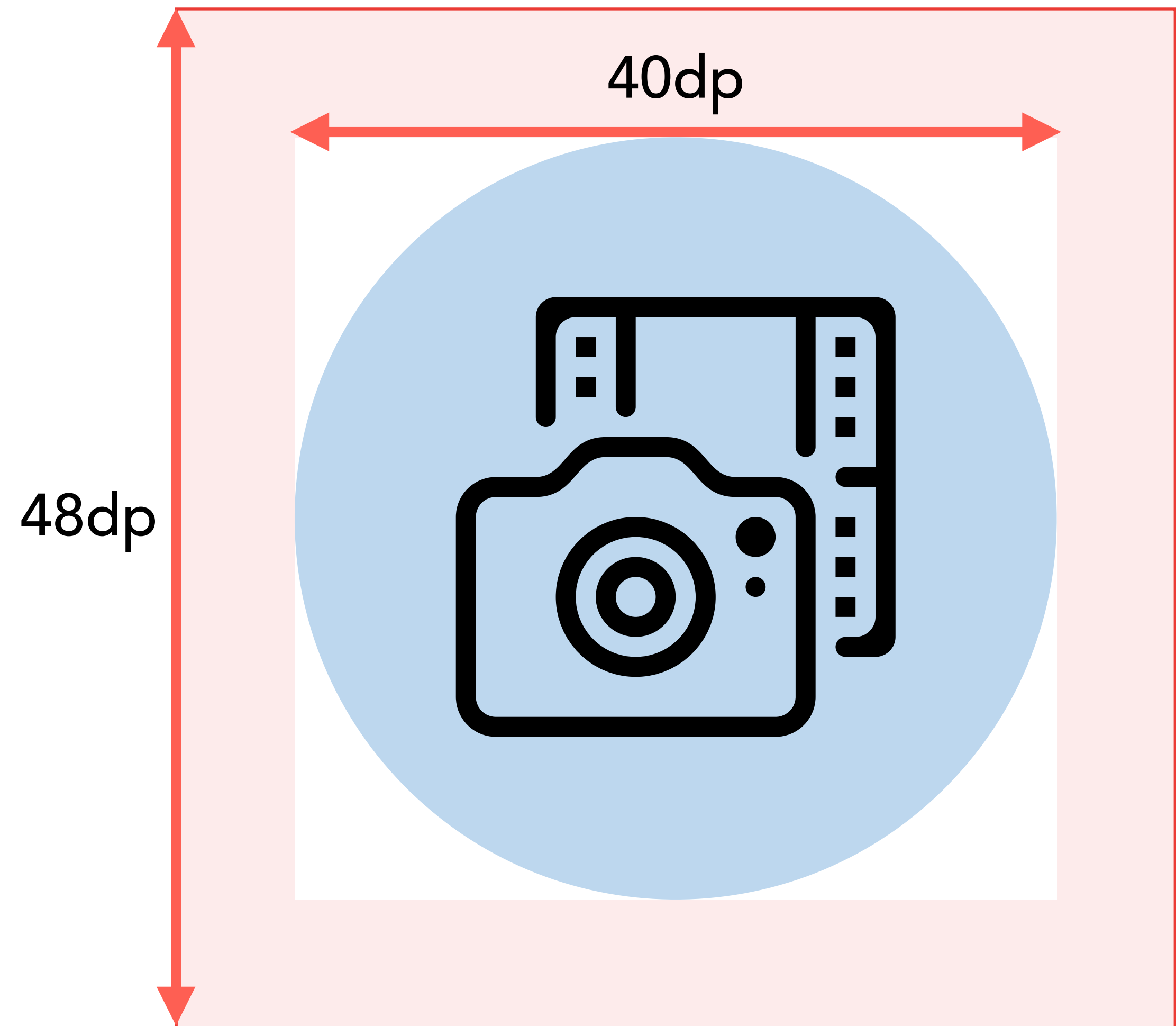
States

	Normal	Hover	Pressed	Selected	Disabled
					
					
					

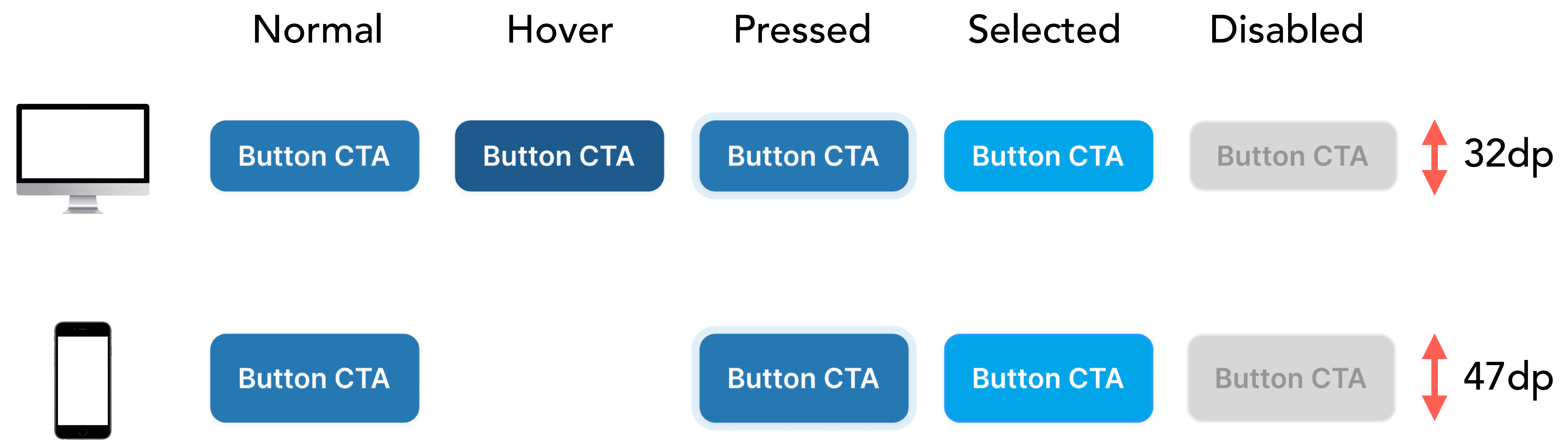
Touch targets

Parts of the display that react to user input

- Often bigger than the visual viewable element e.g. an icon with padding
- 48×48 dp is the minimum touch target size



Touch targets



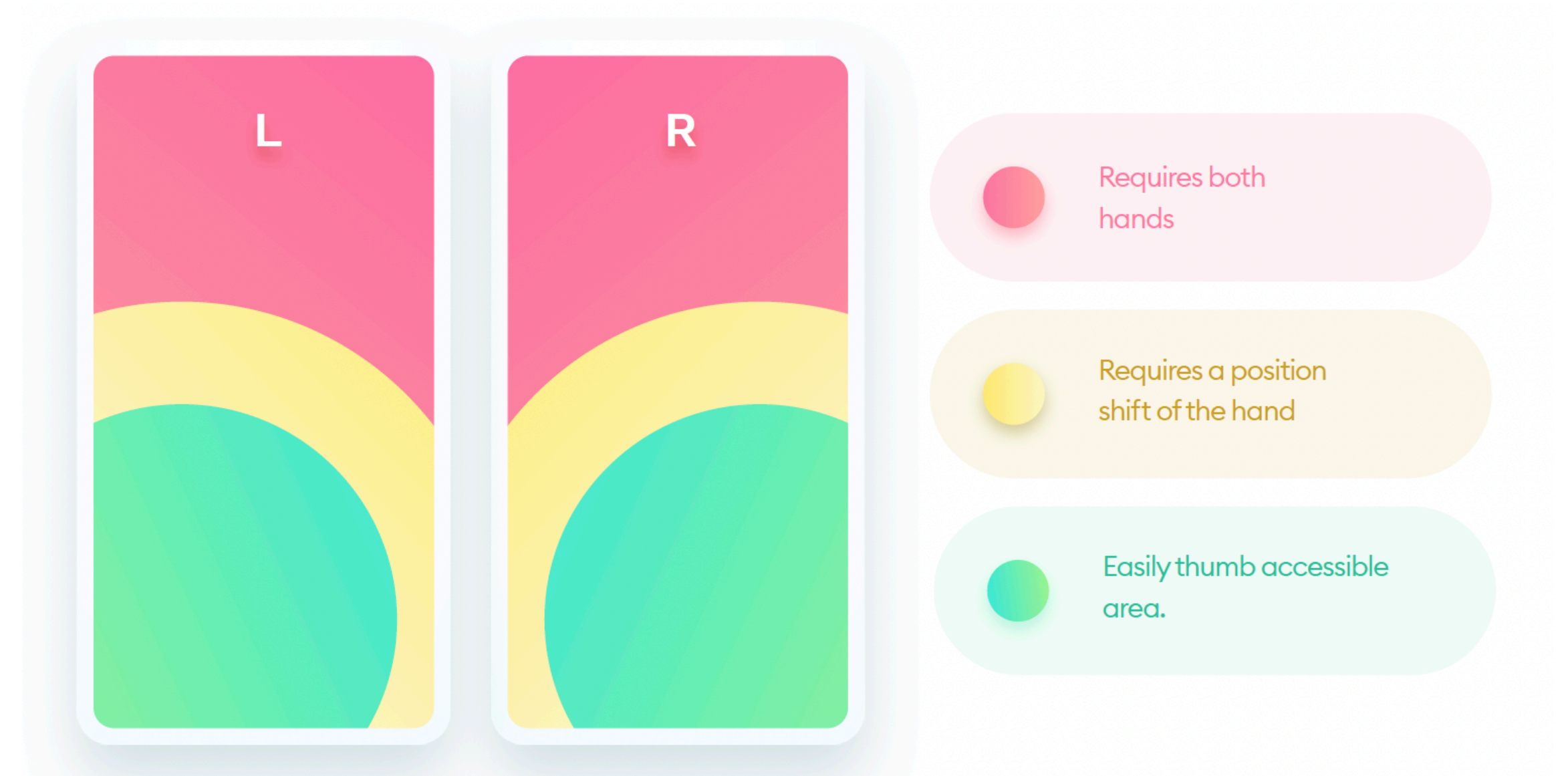
Touch friendly sizes

Touch-Friendly Design

In mobile devices, the average reach is a factor

A randomly pieced together interface can be complicated to use with one hand.

We assume the typical phone usage is a single hand holding the phone with the thumb of the same hand doing most on-screen work.



Tesler's law

For any system there is a certain amount of complexity that cannot be reduced

✓ Also known as the „law of conservation of complexity“

✗ Don't just hide content! Prioritize new



✓ **Progressive disclosure** (interaction design technique) means to display only important actions or content by default. Simplify your design by hiding complexity.

Mobile

Design Patterns

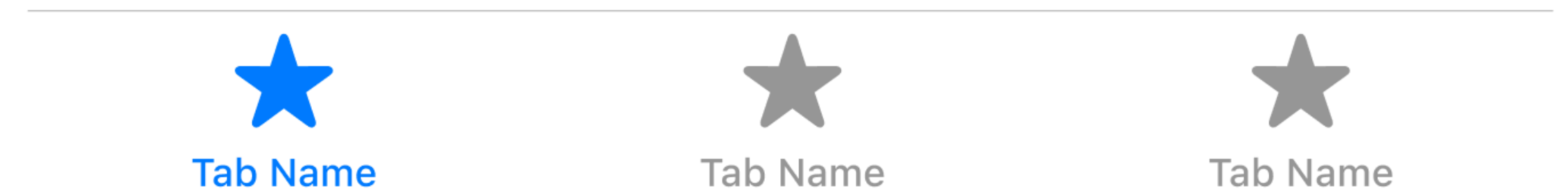
Mobile Design Patterns

- Design Patterns may differ according to device e.g. Small-Scale (Smartwatch) and Large-Scale (Public Display, Kiosk)
- [Breadcrumbs](#) may only appear on desktop
- [Navigation bar](#) may become [hamburger menu](#)



Navigation

- Simple, linear navigation
- Fewer menu levels
- Often visible **tab bar** at the bottom of the screen - and as **horizontal bar** on desktop
- If more than 5 main points are needed, often hidden **hamburger-menu**

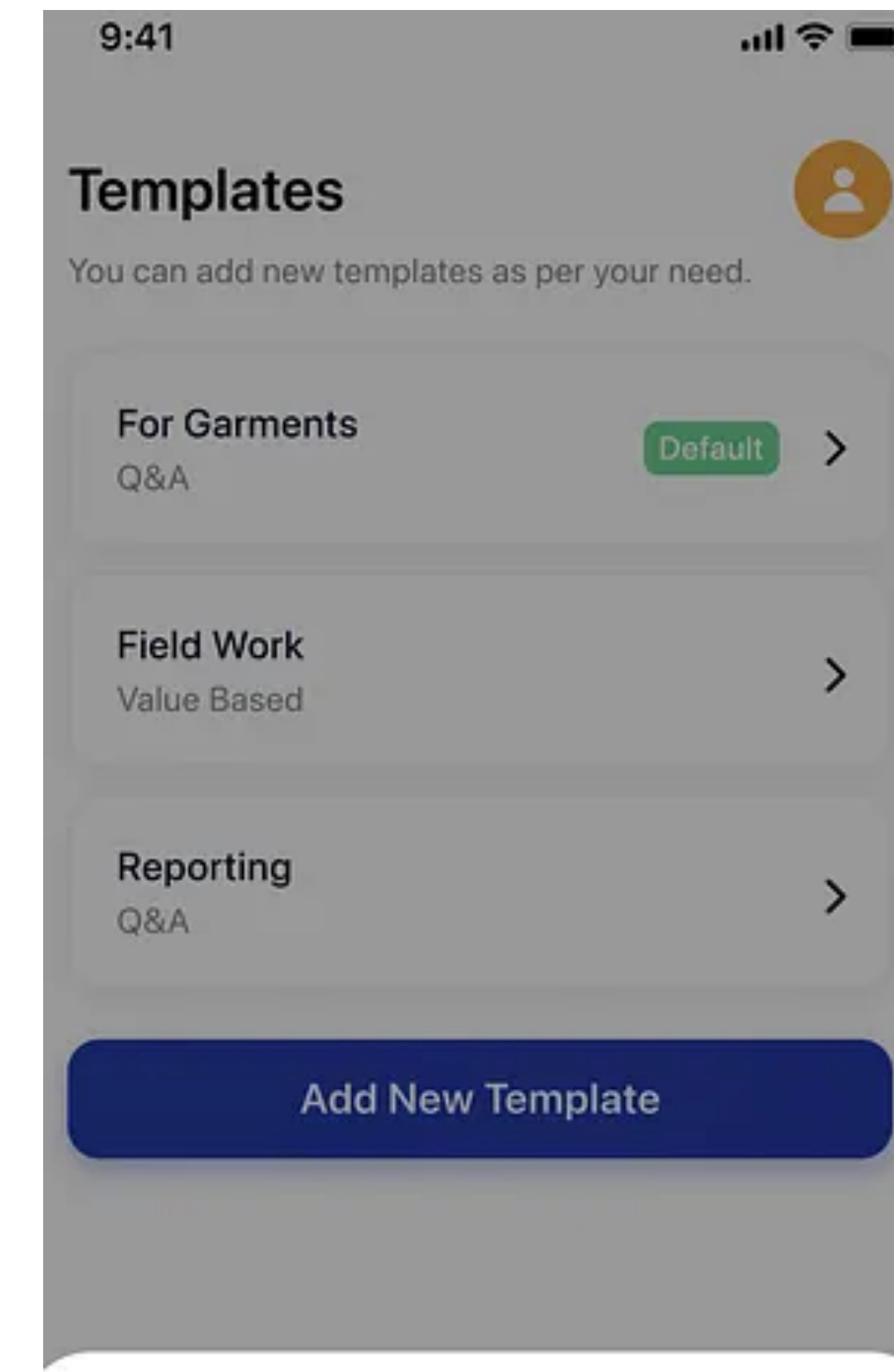


X Don't use tab bar for > 5 destinations

Drop Downs

- Working different than on desktop
- Focuses attention on one task and often hides background
- offers few, clearly marked actions

✓ Thumb friendly



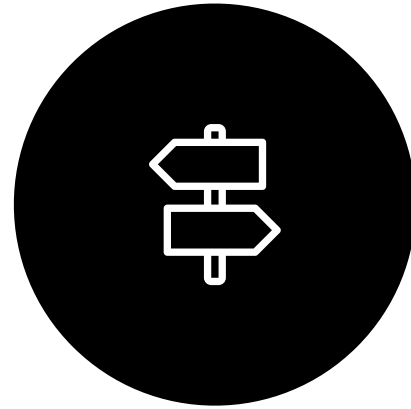
Analyse app

Choose an app you have on your phone

1. 10 minutes to ask following questions:
 - Which responsive patterns do you notice?
 - Which content is priority?
 - Where is progressive disclosure used?
 - Which navigation is used?
2. Note your results
3. Try **figma sites** in a mobile context and insert similar design patterns like your app
4. Each group presents their design and explains:
 - How do patterns change between desktop, tablet and mobile?

UX Management

UX management



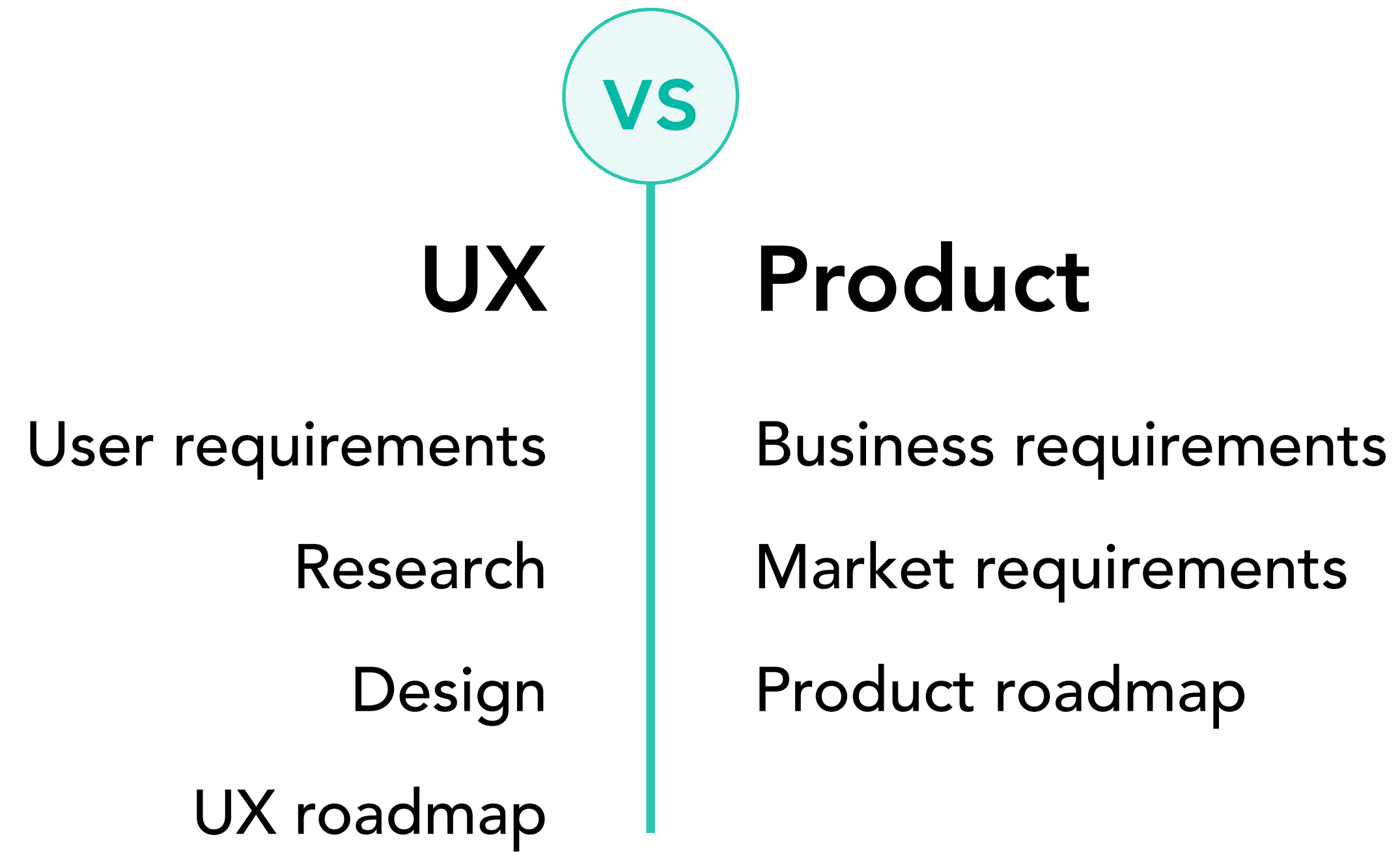
... managing UX across product process to **systematically integrate user needs** into UX strategy, design decisions, agile development and evaluation

Translates **research insights** into user stories, requirements & design decisions

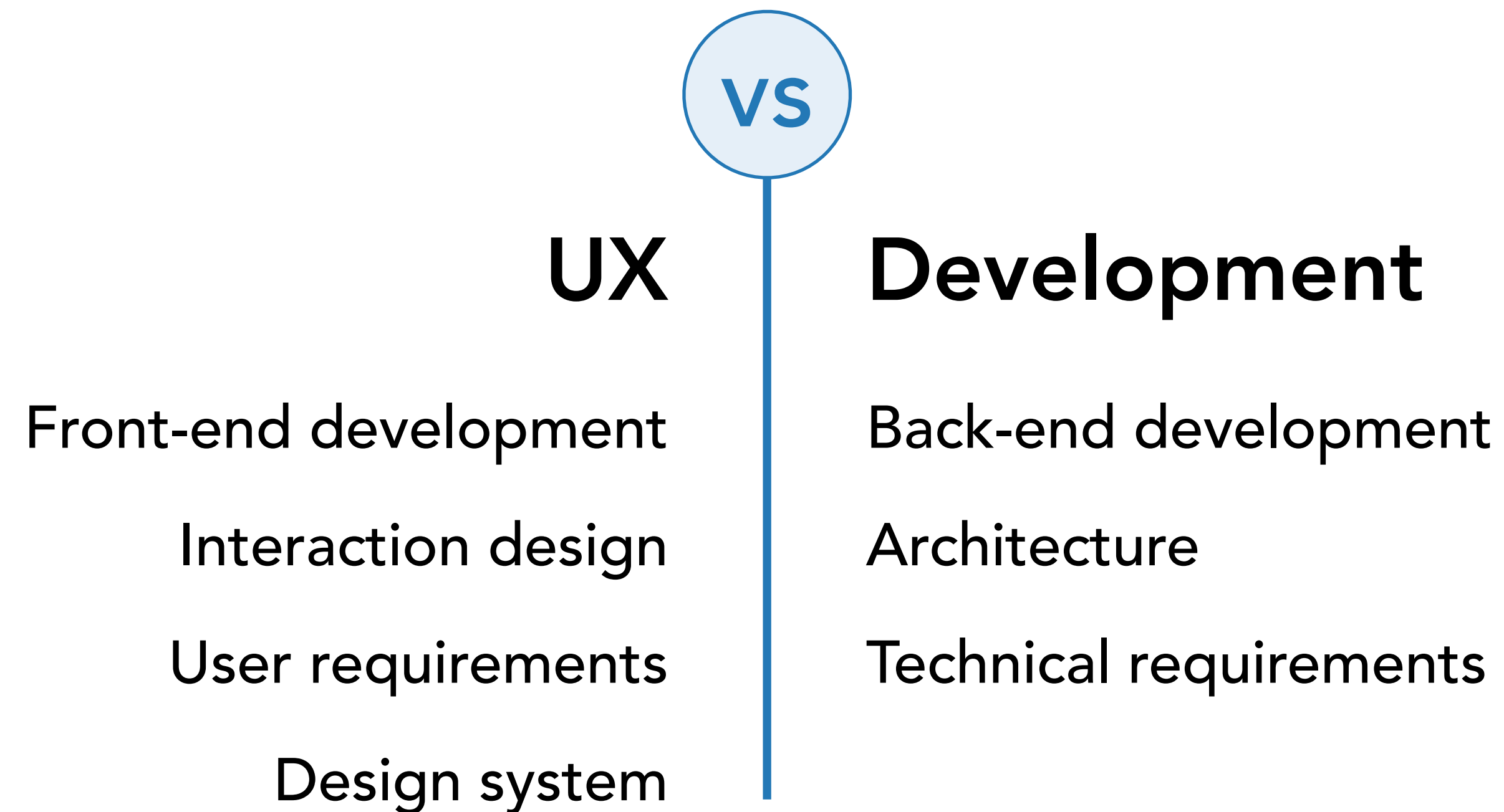
Defines **UX goals, metrics & targets** to evaluate whether the experience works

Connects UX work with **agile** development to consider user needs

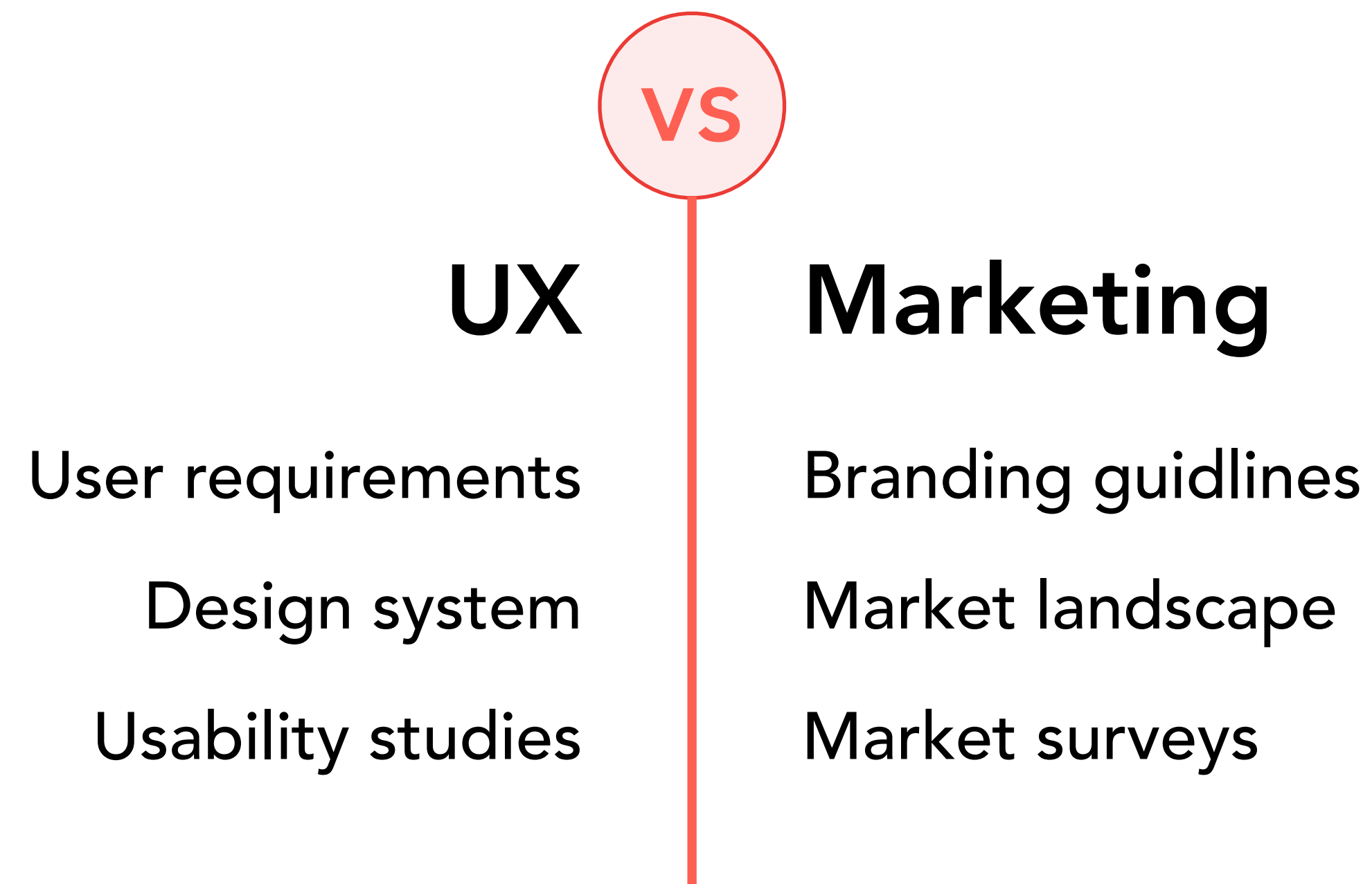
Responsibilities



Responsibilities



Responsibilities

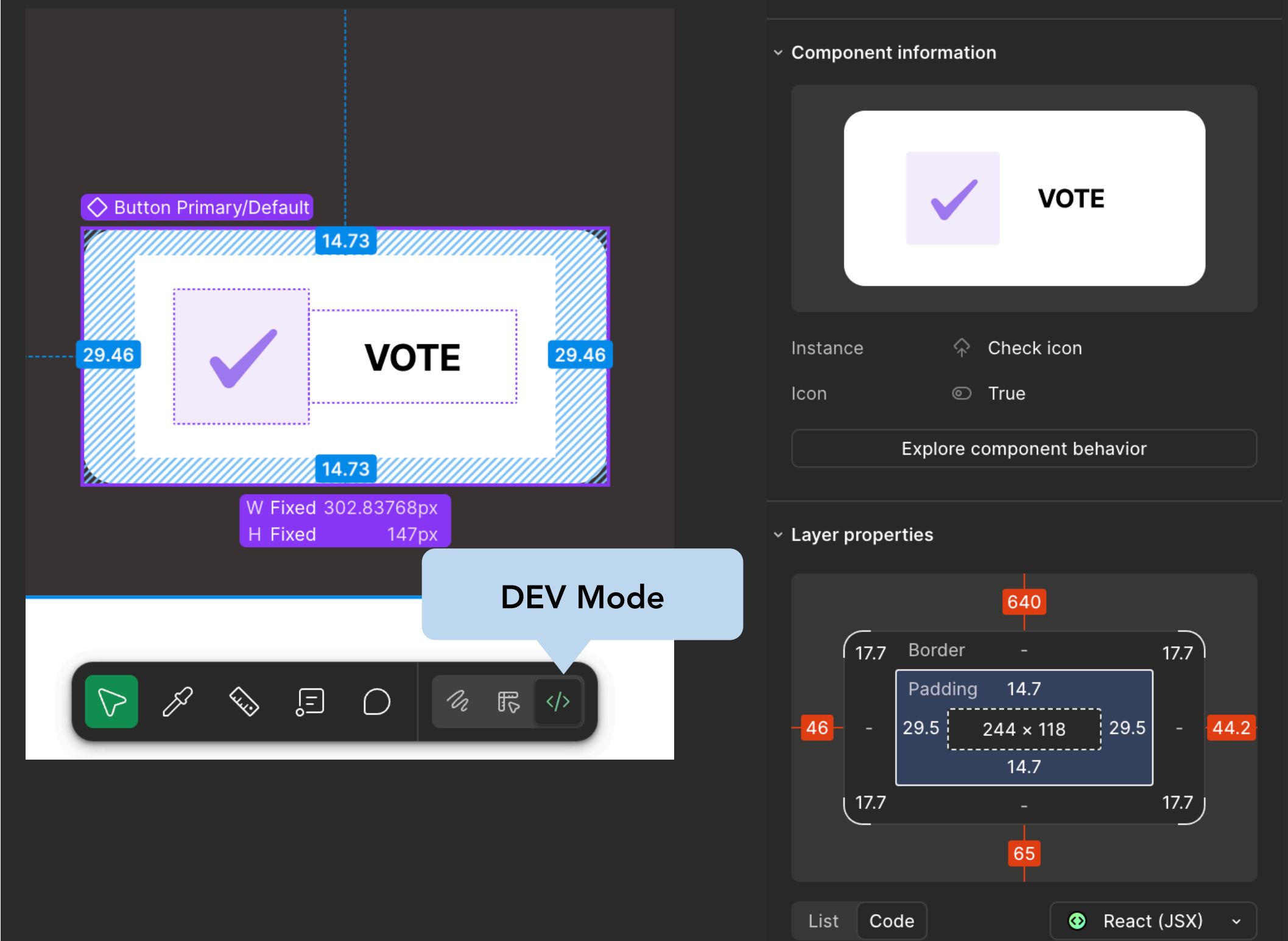


Development <> Design

Challenge: Translating between design and code

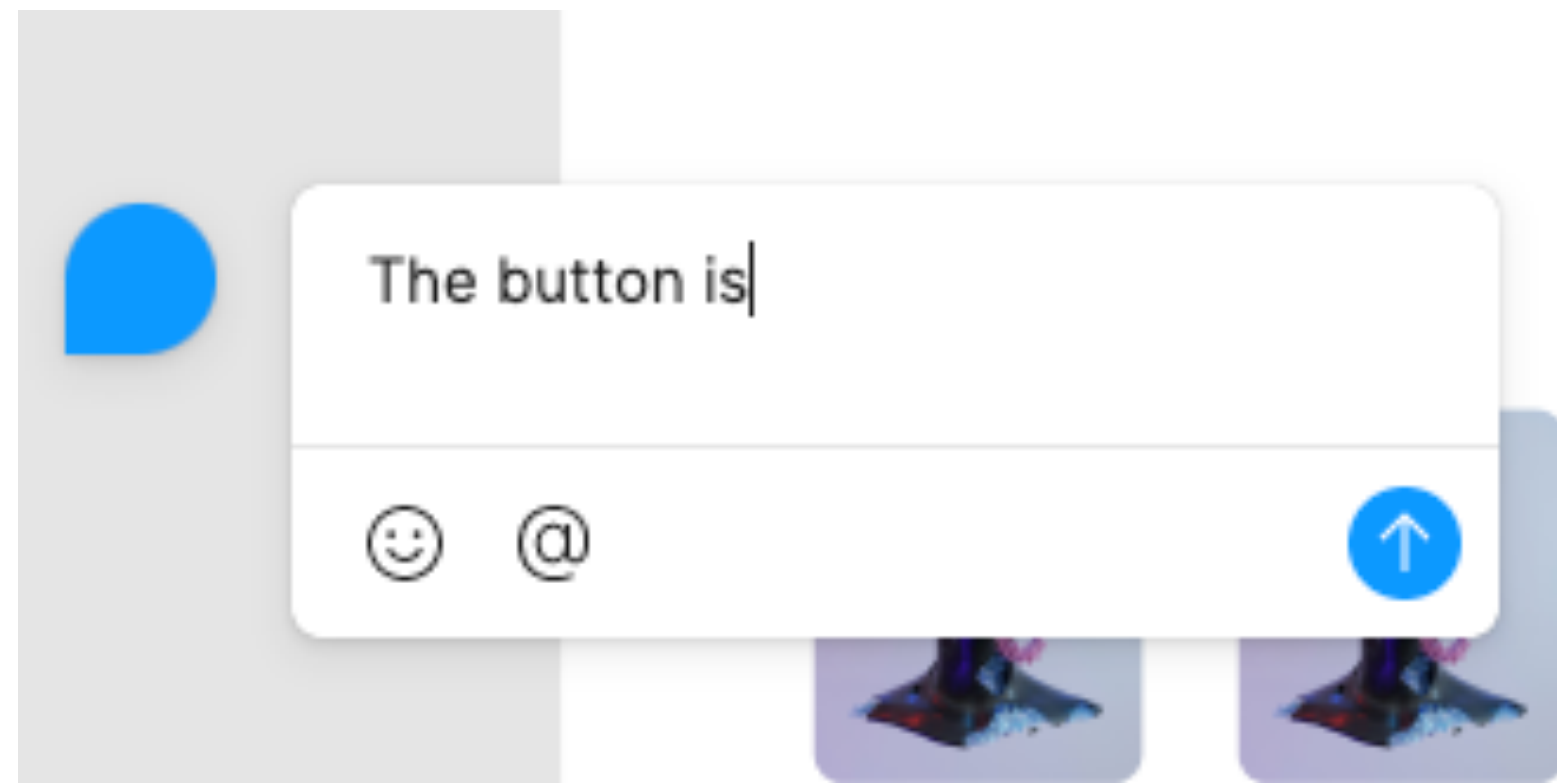
👉 Goal: Fewer misunderstandings, faster implementation

- ✓ Shared components
- ✓ Teach developers measurements & component states
- ✓ Tools for linking (e.g. design ↔ code) or/and storybook
- ✓ Defined tokens and fonts
- ✓ Good design documentation



Development <> Design

- ✓ **Export** layers as PNG, JPG, SVG, PDF
- ✓ **Code exports** (CSS, Angular, React) for frontend development possible with figma
- ✓ **Comments** for team collaboration



Code

CSS ▾



```
/* Logged out */
```

```
position: relative;  
width: 375px;  
height: 812px;
```

```
background: #FFFFFF; 
```

Export



1x

Suffix

PNG ▾



Export Logged out

Documentation & Handover

Updated regularly

Explain design decisions and use cases

Define ownership and update process

Act as single source of truth

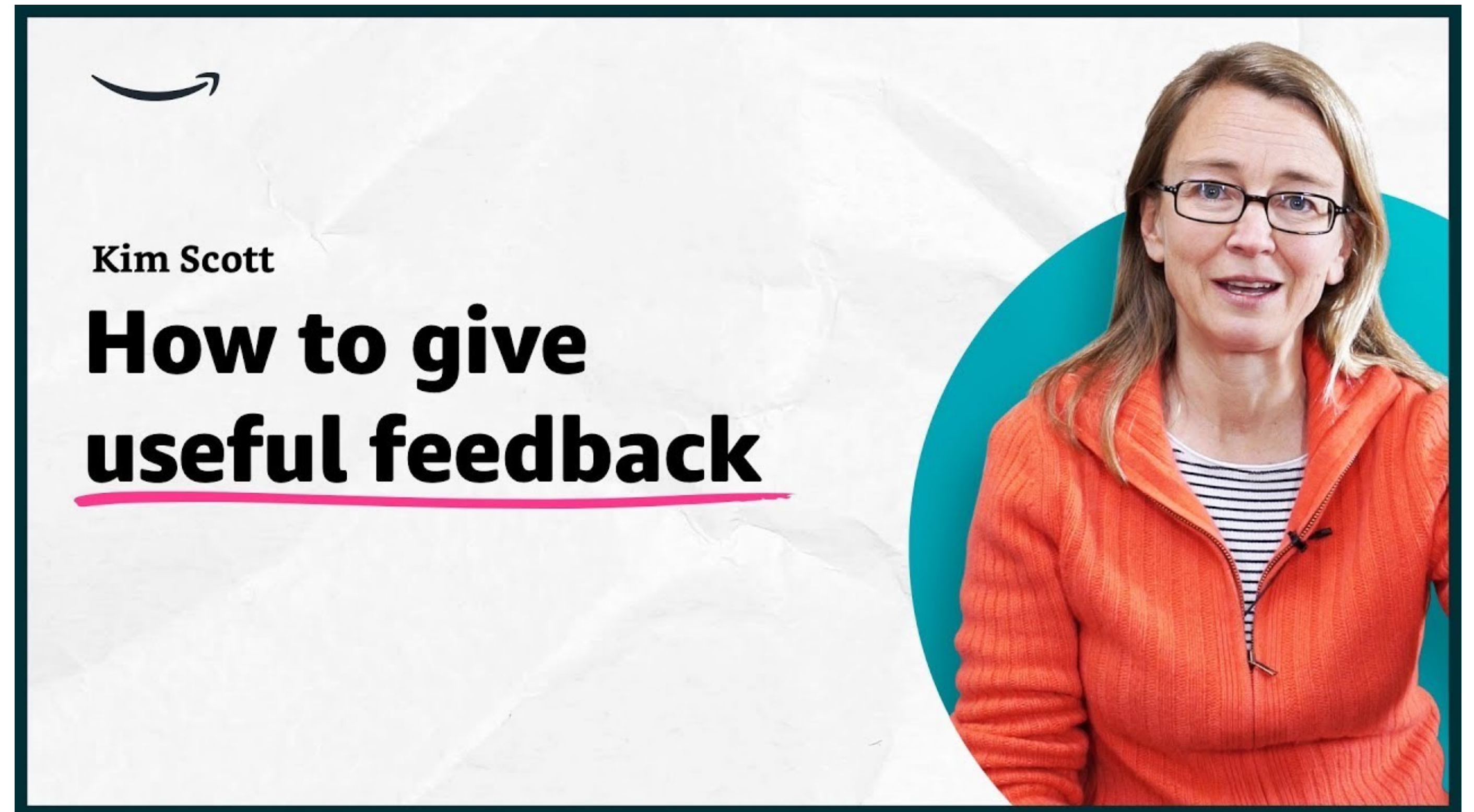
Support consistency and reusing

Include components, code snippets &
design rules

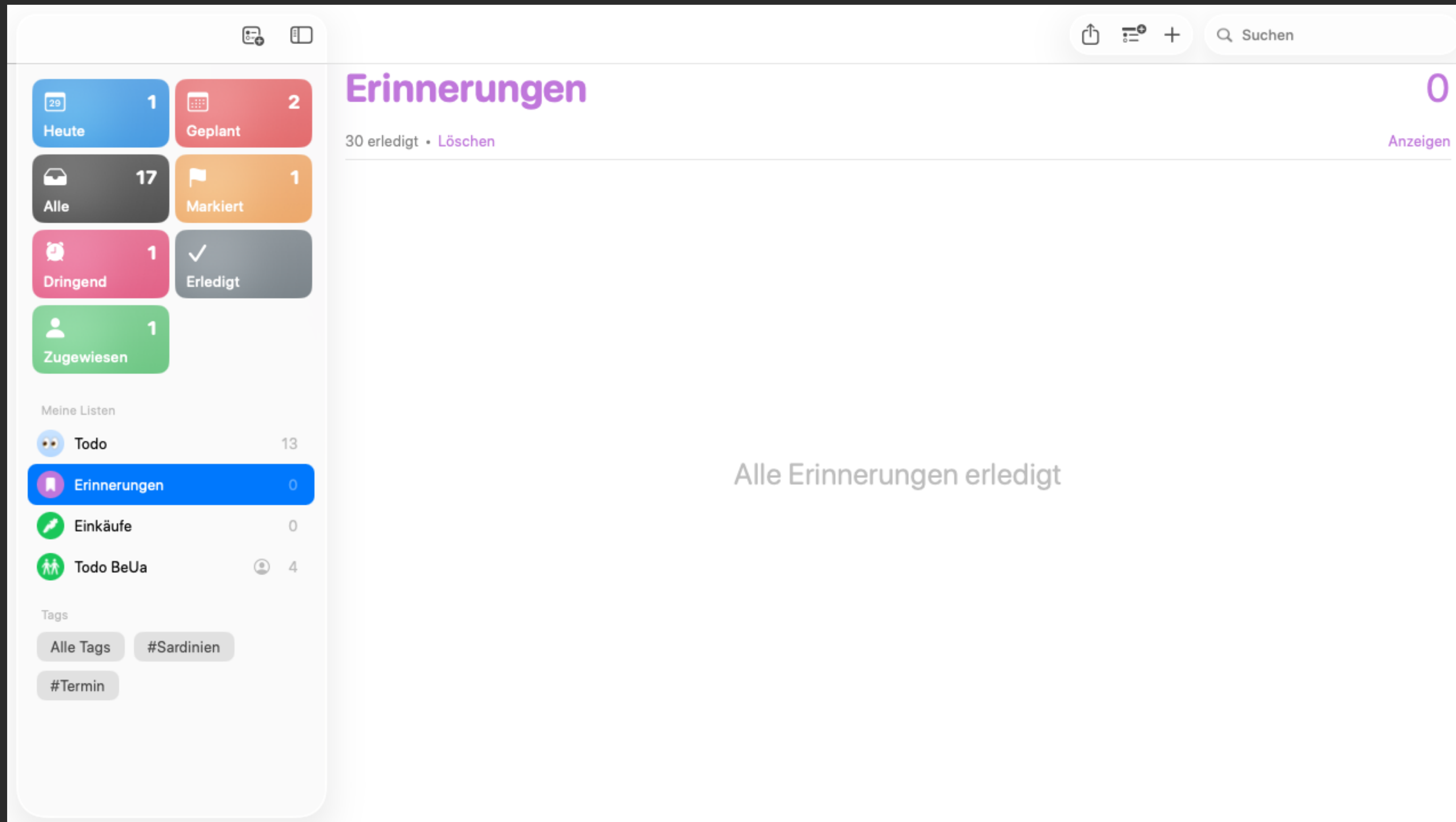
Feedback

- ✓ Specific
- ✓ Include outcome

- X Don't accuse
- X Don't take someone else's interpretation



🤔 Design Critique

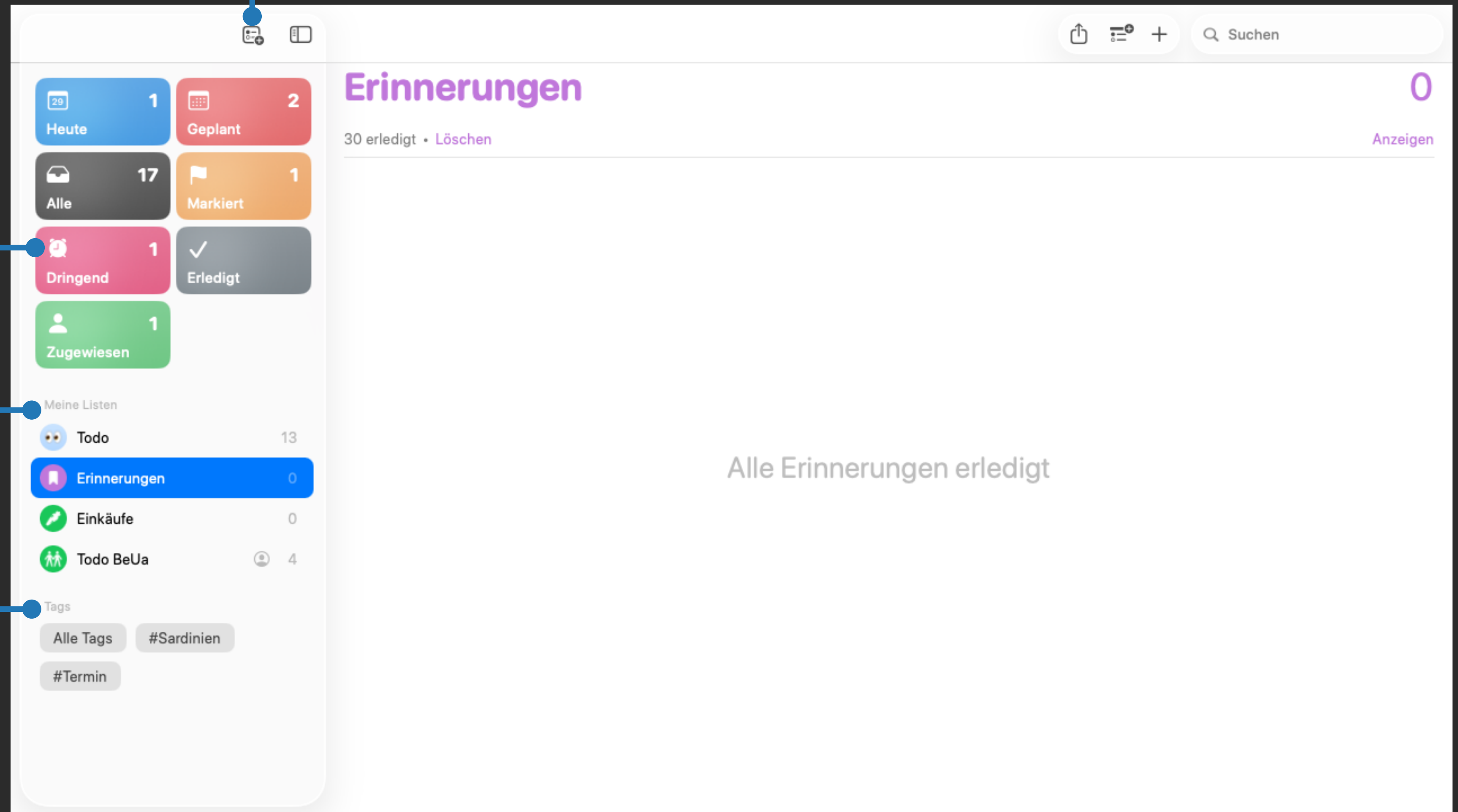


Button not next to list
(proximity)

Not important but
very prominent

Too light
(Accessibility?)

Why do I need
tags?



References

Hartson, R & Pyla, P 2019, *The UX book: Agile UX design for a quality user experience*, Elsevier Inc, Cambridge, MA, Science Direct

Malewicz, M., & Malewicz, D. (2023). *Designing user interfaces*. Hype4.

Tidwell, J., Brewer, C., & Valencia, A. (2020). *Designing interfaces: Patterns for effective interaction design* (3rd ed.). O'Reilly Media.

Twarog, A., & Moller, G. (2024). *Design for developers*. Independently published.

Wathan, S., & Schoger, A. (2018). *Refactoring UI*. Refactoring UI.

Yablonski, J. (2020). *Laws of UX: Using psychology to design better products & services*. O'Reilly Media.